

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
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Date:	31-07-2026	Duration:	60 minutes

Code of Conduct

I A.Akshaya(192425170) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya

ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION MANAGEMENT SYSTEM

1. Identify the Project Vision and Stakeholders

- Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision and Objectives

The vision of the **Academic Project Allocation, Monitoring and Evaluation Management System** is to develop a smart and transparent platform that allocates academic projects based on students' interests, skills, and technical knowledge instead of only considering their CGPA. The system aims to improve learning, project quality, and overall student performance. It also provides an efficient way for faculty members and management to monitor and evaluate projects.

The main objectives of the system are:

- Allocate projects based on students' interests and technical skills.
- Improve practical learning and problem-solving abilities.
- Monitor project progress regularly.
- Ensure fair and transparent project evaluation.
- Reduce manual work in project management.
- Help students develop skills for better career opportunities.

Key stakeholders involved in the system are:

- **Students** – Select interests, receive projects, and submit project work.
- **Faculty Guides** – Guide students, monitor progress, and evaluate projects.
- **Project Coordinator** – Allocates projects, assigns guides, and manages deadlines.
- **Head of Department (HOD)** – Supervises and approves project activities.
- **College Management** – Reviews reports and monitors overall project performance.
- **System Administrator** – Maintains the application, user accounts, and database.
- **External Examiner** – Conducts the final evaluation and awards marks.

2. Create Epics and User Stories

- Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.

Epic 1: User Registration and Login

- As a **student**, I want to register and log in to the system, so that I can access my project details.
- As a **faculty member**, I want to log in securely, so that I can monitor and evaluate student projects.
- As an **administrator**, I want to manage user accounts, so that only authorized users can access the system.

Epic 2: Student Profile and Interest Management

- As a **student**, I want to enter my area of interest, so that I receive a project related to my preferred domain.
- As a **student**, I want to select the programming languages I know, so that suitable projects can be recommended.
- As a **faculty member**, I want to view students' interests and skills, so that I can guide them effectively.

Epic 3: Project Allocation

- As a **project coordinator**, I want to allocate projects based on students' interests and skills, so that project assignment is fair.
- As a **student**, I want to view my allocated project, so that I can start working on it.
- As an **HOD**, I want to approve project allocations, so that the allocation process is transparent.

Epic 4: Project Monitoring

- As a **student**, I want to upload weekly project progress, so that my faculty can review my work.
- As a **faculty member**, I want to monitor project progress, so that I can provide timely guidance.
- As a **project coordinator**, I want to track project completion, so that deadlines are maintained.

Epic 5: Project Evaluation

- As a **faculty member**, I want to evaluate student projects, so that marks are awarded fairly.

- As an **external examiner**, I want to review the final project, so that I can provide final marks and feedback.
- As a **student**, I want to view my evaluation results, so that I understand my performance.

Epic 6: Reports and Notifications

- As a **management member**, I want to generate project reports, so that I can monitor overall project performance.
- As a **student**, I want to receive deadline notifications, so that I can submit my work on time.
- As a **faculty member**, I want to receive project review reminders, so that I do not miss evaluation schedules.

Epic 7: System Administration

- As an **administrator**, I want to manage departments, projects, and users, so that the system functions efficiently.
- As an **administrator**, I want to maintain the database securely, so that user information remains safe.
- As an **administrator**, I want to perform backups, so that project data can be recovered if required.

3. Define Acceptance Criteria

- Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

Acceptance criteria define the conditions that must be satisfied before a feature is considered complete and ready for use. They help developers, testers, and stakeholders understand the expected behavior of the system. In the proposed **Academic Project Allocation, Monitoring and Evaluation Management System**, every major user story has clear acceptance criteria. **Given** that a student has successfully registered and logged into the system, **when** the student enters their area of interest, technical skills, and programming languages, **then** the information should be saved successfully in the database. **Given** that the student's profile is complete, **when** the Project Coordinator allocates a project, **then** the allocated project should be displayed in the student's dashboard. **Given** that the student uploads project progress reports, **when** the faculty guide reviews the submission, **then** the faculty should be able to provide feedback and evaluation. **Given** that the project evaluation is completed, **when** the marks are submitted by the faculty and external examiner, **then** the final results should be visible only to authorized

users. **Given** that the HOD or Management requests project reports, **when** the report generation option is selected, **then** the system should generate accurate and up-to-date reports. These acceptance criteria ensure that every feature functions correctly, meets user requirements, and maintains the quality and reliability of the software system.

4. **Prioritize the Product Backlog**

- Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

The product backlog is a prioritized list of all the features and functionalities required to develop the **Academic Project Allocation, Monitoring and Evaluation Management System**. The backlog is organized based on business value, user requirements, and project dependencies.

- The highest priority is given to **User Registration and Login** because all users must access the system securely before using other features.
- The next priority is the **Student Profile and Interest Module**, where students enter their skills, programming languages, and preferred project domains.
- After collecting this information, the **Project Allocation Module** is implemented because it depends on student profiles and is the core functionality of the system.
- The **Faculty Monitoring Module** is developed next to allow faculty members to track project progress, review submissions, and provide guidance.
- The **Project Evaluation Module** follows, enabling faculty and external examiners to assign marks and feedback.
- The **Report Generation and Notification Module** is then implemented to generate project reports and send reminders for deadlines and reviews.
- Finally, the **Administration Module** is developed to manage users, departments, projects, and system maintenance. Prioritizing the backlog in this order ensures that the most important and dependent features are completed first, allowing the system to function efficiently while meeting the needs of students, faculty members, management, and other stakeholders.

5. **Plan the Sprint Backlog**

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

The first sprint focuses on developing the core features of the **Academic Project Allocation, Monitoring and Evaluation Management System**. The highest-priority user stories are selected so that the basic system is ready for use. The development tasks for Sprint 1 are:

- Create the project database and required tables.
- Design the user registration and login pages.
- Implement secure login for Students, Faculty, HOD, Project Coordinator, and Admin.
- Develop the Student Profile module.
- Add a feature for students to select their area of interest.
- Add a feature for students to select programming languages and technical skills.
- Create the Project Allocation module.
- Enable the Project Coordinator to assign projects based on students' interests and skills.
- Develop the Student Dashboard to display allocated project details.
- Develop the Faculty Dashboard to monitor assigned students.
- Implement basic validation for user inputs.
- Connect the frontend with the backend database.
- Test registration, login, profile creation, and project allocation modules.
- Fix any bugs or errors identified during testing.
- Prepare the completed Sprint 1 features for review and demonstration to stakeholders.

6. Estimate Effort Using Story Points

- Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

Story points are used in Agile development to estimate the effort, complexity, and time required to complete each user story. In this project, the **Fibonacci sequence (1, 2, 3, 5, 8, 13)** is used for estimation because it is simple and widely followed in Scrum. User stories with less complexity are assigned fewer story points, while user stories that require more development, testing, and integration are assigned higher story points. These estimates help the development team plan the sprint efficiently and complete the work within the given time.

The estimated story points for the selected user stories are:

User Story	Story Points	Justification
User Registration and Login	3	Simple functionality involving user authentication and database connection.
Student Profile Management	5	Requires collecting and storing student details, interests, and technical skills.
Interest & Skill Selection	3	Simple form with validation and database storage.
Project Allocation	8	Core functionality that involves matching student interests with suitable projects and faculty guides.
Student Dashboard	5	Displays project details, status, and progress information.
Faculty Dashboard	5	Allows faculty to monitor students, review progress, and provide feedback.
Database Integration	8	Requires connecting frontend and backend with proper data handling and security.
System Testing & Bug Fixing	5	Includes testing all developed modules and resolving identified issues.

The **Fibonacci estimation technique** helps the team understand the relative effort required for each task instead of estimating exact hours. User stories with **3 story points** are considered low complexity, **5 story points** represent medium complexity, and **8 story points** indicate high complexity because they involve multiple modules and dependencies. This estimation allows the team to plan Sprint 1 effectively, balance the workload among developers, and deliver the required features within the sprint duration.

7. Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

The main goal of **Sprint 1** is to develop the core functionalities of the **Academic Project Allocation, Monitoring and Evaluation Management System**. This sprint focuses on creating a secure and user-friendly system where students can register, log in, create their profiles, and enter their areas of interest and technical skills. It also aims to develop the basic

project allocation process so that faculty members and project coordinators can begin managing student projects efficiently. By the end of the sprint, the team should have a working prototype with the essential features that can be demonstrated to stakeholders for feedback and further improvements.

The expected sprint deliverables are:

- A fully functional **User Registration and Login Module**.
- Student Profile Module with personal details, interests, and technical skills.
- Feature for selecting preferred project domains and programming languages.
- Secure authentication for Students, Faculty, HOD, Project Coordinator, and Admin.
- Basic **Project Allocation Module**.
- Student Dashboard to display allocated project details.
- Faculty Dashboard to monitor assigned students.
- Connected frontend and backend database.
- Input validation and secure data storage.
- Basic testing of all implemented modules.
- Bug fixes and performance improvements.
- Sprint Review demonstration for stakeholders.
- Updated project documentation and sprint report.

At the end of Sprint 1, the system should be capable of performing the basic project allocation process and managing user information efficiently. The completed features will provide a strong foundation for the next sprint, where advanced functionalities such as project monitoring, evaluation, notifications, report generation, and analytics will be developed. This ensures steady progress toward building a complete and reliable project management system.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are	Vision is unclear or incomplete. Stakeholders are

	stakeholders and their roles are correctly identified.		missing or incorrectly identified.	poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25